Ricardo (Zhicong) Xie

Product / Hardware Engineering Intern | CAD, Embedded C and Mechatronics

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Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical engineering master's student with project evidence across CAD modelling, enclosure design, embedded C, sensor systems and early validation workflow.
- Most relevant work: ruggedized IoT sensor module, dynamic weighing firmware awarded Software Copyright Registration No. 2024SR0141888, SolidWorks ventilation hardware layout and hands-on manufacturing training.
- Project screenshots and source links: https://zxie588-alt.github.io

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Relevant coursework: Advanced Industry 4.0 Smart Manufacturing, Manufacturing and Industrial Systems, Engineering Methods

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Manufacturing workshop training: prepared AutoCAD/DXF laser-cut profiles, compared drawing intent with cut parts, and worked around CNC workflow, workholding, chip formation and shop-floor safety awareness.
- Engineering practice: interpreted force-displacement test output, practised robotic arm control, and acted as a small-group lead explaining automotive seat/interior structures to peers.
- Technical visual communication: former student news-department lead with photography/event documentation experience; professional working English and native Mandarin.

PROJECT WORK

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, validation workflow

- Designed a SolidWorks enclosure design package with screw bosses, gasket stack-up, PCB and battery envelopes, cable/strap mounts and service access.
- Applied DFM principles to improve enclosure features for CNC machining review and 3D-print prototype preparation.
- Added FEA-style screening for corner load, strap lug and screw boss regions; kept an open test list for sealing, impact, thermal exposure and firmware validation.
- Built prototype firmware in Keil C with sensor-driver abstraction, battery monitoring and a simple module state machine; documented the electronics and firmware architecture in the portfolio.

Dynamic Livestock Weighing System - Software Copyright Registered

STM32, HX711, embedded C, Bluetooth/serial communication

- Developed STM32 firmware for pressure-sensor acquisition using HX711 hardware and real-time data transmission to an upper computer; system awarded Computer Software Copyright Registration No. 2024SR0141888.
- Supported dynamic weighing logic for improved signal stability under field-like conditions.

NZ Residential MVHR / ERV Ventilation Design

Native Revit 2026 MEP coordination, SolidWorks, AutoCAD/DXF, HVAC sizing, NZBC G4/H1 context, commissioning plan

- Developed a New Zealand residential MVHR/ERV ventilation design aligned with local building-code and building-services workflow expectations.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Built a native Revit 2026 MEP coordination model with a one-level residential layout, MVHR unit, supply/extract routes, diffusers/grilles, airflow schedule, three portfolio sheets, drawing-review comments and a simple clash-check note.
- Sized the system for 60 L/s balanced supply/extract in normal operation and 85 L/s boost extract mode, calculating external static pressure targets of about 80 Pa and 120 Pa.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; developed a commissioning test plan covering CO2, humidity, airflow and fan-power measurements.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared 2D AutoCAD/DXF geometry for laser cutting and engraving, then compared drawing intent with the physical cut plate.
- Observed and practised CNC machining workflow including workholding, spindle/tool motion, cutting fluid, chip formation and basic workshop safety protocols aligned with general WorkSafe NZ expectations.
- Reviewed machined gear and shaft components to connect CAD features with real manufacturing constraints such as tolerances, edge quality and tool access.

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Developed a shell FEA model of a composite Formula SAE front wing in Abaqus/Standard using S4R elements, layered laminate definitions and local mesh refinement at high-stress attachment points.
- Converted aero downforce/drag, lateral cornering and 4.1 G bump conditions into structural load cases for strength and stiffness checks.
- Evaluated ply-level stress utilisation against fatigue and ultimate failure criteria, achieving about 74% peak utilisation under the governing 4.1 G bump case with SF = 1.5.

TECHNICAL SKILLS

- Proficient in: SolidWorks assemblies, enclosure features, gasket/screw-boss layout, STEP/STL export, component documentation
- Familiar with: STM32 C, Keil 5, Proteus, HX711, sensors, battery-monitor logic, serial/Bluetooth communication, PCB/circuit basics
- Engineering methods: DFM notes, FEA-style screening, test-list writing, firmware state-machine documentation, source-linked portfolio evidence